RESTAURANTOPS AI • SOFTWARE ENGINEERING SPECIFICATION

Software Requirements Specification (SRS)

AI-Powered Restaurant Inventory and Procurement Management System

Document Title	Software Requirements Specification (SRS)
Project Name	RestaurantOps AI
Document Version	1.0
Document Status	Reconstructed Project Specification • Submitted for Review
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Date of Preparation	November 2026
Target Industry	Restaurant, Culinary, and Hospitality
Functional Role	AI Inventory and Procurement Manager
Target Environment	Single-Restaurant MVP (Web SPA & REST API)

Specification Notice & Formal Scope:
This document provides the complete, reconstructed Software Requirements Specification (SRS) for the RestaurantOps AI system, capturing all functional requirements (FR-01 to FR-08), architectural layers, data flows, integration contracts, and operational criteria established during project initialization and core backend development. It serves as the baseline for system verification and final approval.

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1. Introduction

The restaurant and food service industry is characterized by razor-thin operating margins, extreme demand volatility, and heavy reliance on highly perishable ingredients. Managing kitchen inventory directly dictates profitability: stockouts lead to immediate lost revenue and frustrated diners, while over-purchasing results in food spoilage, tied-up working capital, and elevated Cost of Goods Sold (CoGS).

Traditionally, culinary inventory control has depended on manual clipboard counts, physical checklists, and disconnected spreadsheets. These methods suffer from human calculation error, lack real-time visibility, and fail to anticipate impending shortages before dinner service. **RestaurantOps AI** is engineered as a modern, full-stack, AI-powered inventory and procurement management solution. By binding point-of-sale customer orders to granular ingredient consumption via recipes, the system delivers continuous stock visibility, proactive risk alerting, and conversational intelligence powered by Google's Gemini API.

2. Purpose

The primary purpose of RestaurantOps AI is to provide restaurant owners, executive chefs, and inventory managers with an intelligent, automated platform that eliminates operational guesswork and streamlines daily pantry operations.

- **Automate ingredient depletion:** Automatically deduct precise ingredient portions upon every menu item sale through configured recipe bill-of-materials.
- **Eliminate ordering guesswork:** Generate deterministic reorder recommendations calculated from actual sales velocity and minimum stock thresholds.
- **Mitigate spoilage risk:** Continuously monitor ingredient shelf life and flag batches approaching expiration at least 48 to 72 hours in advance.
- **Provide AI-driven operational assistance:** Deliver natural language inventory insights and trend explanations through an integrated Gemini AI agent.
- **Enforce governance:** Maintain a strict 'Human-in-the-Loop' workflow requiring managerial review and explicit approval for all purchasing proposals.

3. Problem Statement

Independent and commercial dining establishments grapple with systemic inefficiencies across pantry and supply chain workflows:

- **Manual & Error-Prone Tracking:** Pantry audits conducted late at night via paper logs suffer from arithmetic inaccuracies and inconsistent logging, obscuring actual stock positions.
- **Unexpected Mid-Service Stockouts:** Exhaustion of key ingredients during peak hours forces kitchens to '86' signature dishes, directly damaging revenue and guest loyalty.
- **Overstocking & Locked Capital:** Buffer purchasing driven by fear of stockouts ties up essential operating cash and creates physical overcrowding in cold and dry storage.
- **Perishable Waste & Spoilage:** Failure to track batch expiration dates leads to spoiled proteins, dairy, and produce, driving food waste expenses up to 4-10% of total food costs.
- **Fragmented Procurement:** Orders placed haphazardly across phone calls and vendor portals lack alignment with current stock levels or historical consumption.
- **Lack of Actionable Insights:** Managers lack synthesized visibility combining recent sales velocity, storage status, and ambient context into unified operational action plans.

4. Proposed Solution

RestaurantOps AI resolves these challenges through a unified, service-oriented architecture combining automated inventory math with advanced AI reasoning:

- **Recipe-Driven Depletion:** Every recorded sale initiates an atomic deduction across all associated ingredients, preserving an exact digital replica of kitchen stock.
- **Immutable Transaction Ledger:** All stock modifications (sales depletions, manual reconciliations, waste records, restocks) are recorded to an auditable transaction journal.
- **Deterministic Procurement Logic:** Reorder quantities are computed mathematically using strict threshold formulas: $Q = \max((2 \times \text{min}) - \text{current}, \text{min})$.
- **AI Contextual Analysis:** A Gemini-powered agent translates stock deficiencies into human-readable procurement recommendations with accurate cost estimates.
- **Human-in-the-Loop Governance:** The MVP requires explicit manager review and approval before purchase recommendations transition to actionable purchasing orders.
- **Automated Proactive Alerts:** Background APScheduler workers sweep stock levels periodically and transmit urgent low-stock and near-expiry alerts via Telegram.

5. Project Objectives

ID	Strategic Objective	Target Metric / Acceptance Threshold
OBJ-01	Automated Recipe Consumption	100% of recorded sales automatically deduct linked ingredients in an atomic database transaction.
OBJ-02	Real-Time Inventory Visibility	Pantry balances update with < 200 ms latency; dashboard reflects actual database state immediately.
OBJ-03	Early Shortage Detection	Zero untracked stockouts; 100% of items at or below minimum threshold flagged automatically.
OBJ-04	Perishable Waste Mitigation	Identify 100% of ingredients expiring within 7 days, highlighting items with < 48 hours remaining.
OBJ-05	Procurement Efficiency	Reduce time required for managers to draft daily vendor orders by at least 60%.
OBJ-06	Grounded AI Assistance	Gemini agent answers user inquiries using validated backend tools with 0 direct SQL exposure.
OBJ-07	Complete Movement Auditability	Maintain an immutable audit trail for 100% of stock additions, depletions, and reconciliations.

6. Target Users and Stakeholders

RestaurantOps AI supports five operational personas within the restaurant management hierarchy:

Stakeholder Role	Core Responsibilities	Primary System Interactions
Restaurant General Manager	Daily operations, P&L; oversight, purchasing authorization, team management.	Executive Dashboard, Procurement Approval/Rejection, Telegram Alerts, AI Assistant.
Kitchen / Inventory Manager	Pantry audits, physical stock receiving, recipe verification, culinary quality.	Inventory Grid, Stock Adjustments, Expiry Tracker, Menu & Recipe Studio.
Procurement Personnel	Vendor communication, supplier pricing negotiations, purchase order dispatch.	Approved Purchase Orders, Reorder Summary Reports, Vendor Spend Summaries.
Restaurant Owner / Executive	High-level profitability, food cost percentage (CoGS), multi-period margin analysis.	Dashboard Analytics, Historical Consumption Reports, Waste Exposure Trends.
System Administrator	System configuration, security credentials, scheduled job monitoring, uptime.	Health Check API, Automation Execution Logs, Database Migration Commands.

7. Functional Requirements

The functional scope of RestaurantOps AI is organized into eight formal requirement blocks (FR-01 to FR-08). Each requirement defines its operational capability, technical constraints, and verifiable acceptance conditions.

FR-01: Inventory Management

- **Description:** The system shall provide comprehensive CRUD capabilities for restaurant ingredients.
- **Attributes Tracked:** Ingredient name, unique SKU, culinary category (Proteins, Dairy, Produce, Dry Goods, Bakery, Beverages, Seasonings), measurement unit (kg, g, l, ml, piece), current stock balance, minimum threshold (`min_quantity`), cost per unit, and optional expiration date.
- **Business Rules:** Stock quantities must be non-negative (`current_quantity >= 0`). Deletion of ingredients referenced by active recipes or sales history is prohibited.
- **Acceptance Conditions:** Ingredient creation succeeds with valid data; duplicate SKUs return HTTP 400; category filtering and full-text name/SKU search function correctly.

FR-02: Sales and Ingredient Consumption Tracking

- **Description:** The system shall record restaurant customer sales, associate menu items with recipes, and automatically deduct underlying ingredients.
- **Recipe Bill-of-Materials:** Menu items are linked to ingredients via `recipe_ingredients` specifying exact `quantity_required` per portion.
- **Depletion Process:** On sale recording, the backend retrieves all item recipes, aggregates total consumption, validates stock availability, and deducts quantities atomically.
- **Acceptance Conditions:** Selling 2 Chicken Burgers deducts 0.30 kg chicken, 2 buns, and 2 cheese slices in an atomic transaction; insufficient stock blocks the sale cleanly.

FR-03: AI-Powered Inventory Analysis

- **Description:** The system shall analyze inventory levels, consumption velocity, and sales trends to identify anomalies and operational risks.
- **Analytical Capabilities:** Classify ingredients into `Critical`, `Low Stock`, and `Healthy`; calculate burn rates based on recent sales; generate natural language summaries.
- **Acceptance Conditions:** Low-stock ingredients are identified deterministically; Gemini generates clear, evidence-based commentary contextualizing stock variances.

FR-04: Procurement Planning

- **Description:** The system shall generate deterministic procurement recommendations and manage their review and approval lifecycle.
- **Reorder Calculation:** Suggested replenishment quantity: $Q = \max((2 \times \text{min_quantity}) - \text{current_quantity}, \text{min_quantity})$.
- **Order Lifecycle:** Order proposals transition from `draft` to `approved` or `rejected`. Total estimated purchase cost is computed using unit costs.
- **Acceptance Conditions:** Recommendations display itemized quantities and estimated totals; manager approval updates order status and timestamps; no automated supplier dispatch occurs in MVP.

FR-05: Expiration and Waste Monitoring

- **Description:** The system shall monitor batch expiration dates to identify ingredients at risk of spoilage and calculate financial waste exposure.

- **Horizon Classification:** Flag items into `Expired`, `Urgent (< 48 hours)`, `Approaching (< 7 days)`, and `Safe`.
- **Acceptance Conditions:** The expiring items endpoint accurately filters ingredients within the requested lookahead window; at-risk dollar value is computed accurately.

FR-06: Notifications and Reporting

- **Description:** The system shall transmit proactive operational alerts and daily summaries to restaurant managers via Telegram.
- **Notification Types:** Instant low-stock alerts, near-expiry batch warnings, and a consolidated morning daily operational report.
- **Acceptance Conditions:** Messages are dispatched asynchronously using `httpx`; failed network requests log warnings to `automation_logs` without failing core APIs.

FR-07: AI Assistant

- **Description:** The system shall provide an interactive conversational AI interface powered by Google Gemini and the official `google-genai` SDK.
- **Tool-Calling Architecture:** Gemini interacts with the system exclusively through registered backend tool declarations mapped to service-layer methods.
- **Acceptance Conditions:** The AI retrieves real inventory data via service tools; direct raw SQL querying by the model is blocked; answers are strictly grounded in restaurant data.

FR-08: Scheduled Automation

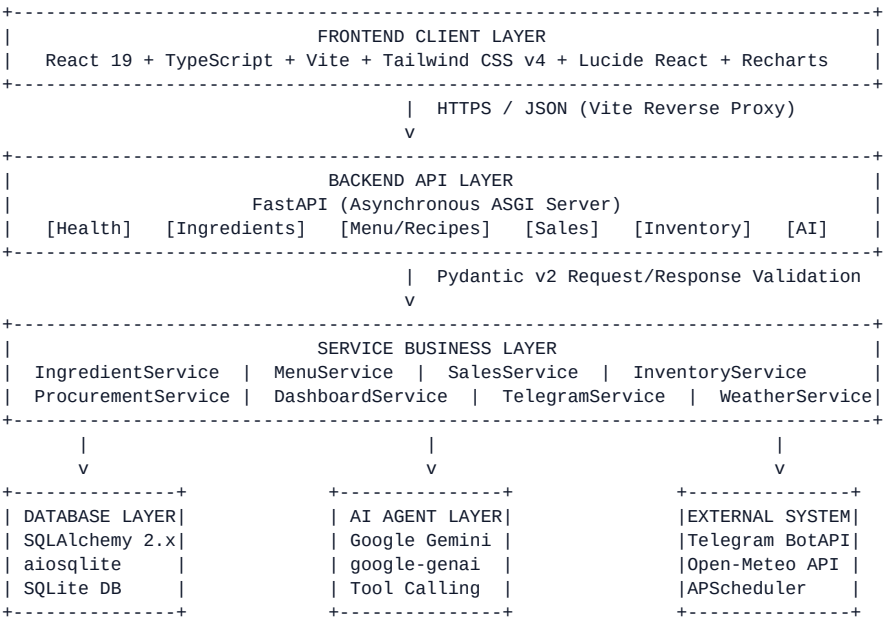
- **Description:** The system shall run background scheduled routines using APScheduler to execute routine checks and audit reporting.
- **Automated Tasks:** Periodic inventory health sweeps, automated low-stock detection, and scheduled morning report generation.
- **Acceptance Conditions:** Scheduled jobs run at configured cron intervals without user intervention; execution results, duration, and errors are written to `automation_logs`.

8. Non-Functional Requirements

Category	Specification & Technical Criteria
Performance	Core CRUD and inventory query endpoints must respond in < 200 ms under typical single-restaurant concurrent loads. Database queries must utilize indexes on SKUs, categories, and foreign keys.
Security & Isolation	All external credentials (GEMINI_API_KEY, TELEGRAM_BOT_TOKEN) must reside strictly on the backend in .env and must never be transmitted to the frontend bundle.
Transaction Atomicity	Sales recording, ingredient depletions, and audit logging must execute within an atomic transaction. Any runtime error triggers an immediate database rollback.
Maintainability	Strict layered architecture (Router $\rightarrow$ Service $\rightarrow$ Model). Pydantic v2 schemas for all I/O contracts. Alembic for database version control and migration management.
Scalability	Stateless FastAPI architecture running asynchronously on uvicorn. Clean SQLAlchemy ORM abstraction enables frictionless transition from SQLite to PostgreSQL in multi-unit scaling.
Usability & Responsiveness	Responsive web interface styled with Tailwind CSS v4, supporting desktop terminals, kitchen tablets, and mobile devices. Conforms to WCAG 2.1 AA accessibility standards.
Data Integrity	Database constraints enforce non-negative stock (current_quantity >= 0), unique SKU keys, and foreign-key referential integrity across recipes and sales records.
Standardized Error Handling	All responses conform to a unified JSON response envelope (ApiSuccess<T> or ApiError), providing structured error messages and machine-readable error reasons.

9. System Architecture

RestaurantOps AI adheres to a strict layered architecture with clear separation of responsibilities. The service layer acts as the single gated point of entry for all business logic, shielding the database from direct manipulation by web controllers, background tasks, or AI agents.



- **Single Gated Service Layer:** All database operations flow through dedicated service functions. AI agents and background jobs call the same validated methods as web controllers.
- **End-to-End Asynchronous I/O:** FastAPI, aiosqlite, and SQLAlchemy 2.x async ORM ensure non-blocking event loop execution across all database queries and HTTP calls.
- **Type-Safe Contract Enforcement:** Pydantic v2 models validate all incoming payloads and serialize all outgoing JSON envelopes, preventing malformed data transmission.

10. Third-Party API Integrations

External Service	SDK / Protocol	Functional Responsibility & Architectural Boundaries
Google Gemini API	google-genai REST / gRPC	Powers conversational assistant, inventory anomaly analysis, and procurement drafting. Interacts via registered service tool declarations; direct database querying is strictly prohibited.
Telegram Bot API	httpx HTTPS REST	Outbound notification pipeline for urgent low-stock alerts and morning operational digests. Operates asynchronously; failures log to database without interrupting user transactions.
Open-Meteo Weather	httpx HTTPS REST	Fetches local temperature and precipitation probabilities to provide contextual operational footfall insights. *Weather data is strictly contextual and never used as a primary inventory depletion signal.*

11. System Workflows

RestaurantOps AI coordinates ten primary end-to-end operational workflows spanning daily restaurant operations:

- **1. Inventory Item Onboarding:** Manager inputs ingredient specifications (name, SKU, category, unit, cost, min_quantity) $\rightarrow$ Pydantic validates payload $\rightarrow$ Database persists record and initializes stock balance at zero or intake

level.

- **2. Menu Item & Recipe Configuration:** Manager creates a menu item $\rightarrow$ specifies linked recipe ingredients and exact unit portions $\rightarrow$ Backend validates existence of all referenced ingredients $\rightarrow$ Persists relational recipe mapping.
- **3. Point-of-Sale Recording & Atomic Depletion:** POS records dish sale $\rightarrow$ SalesService retrieves recipes $\rightarrow$ verifies sufficient stock $\rightarrow$ deducts ingredients $\rightarrow$ writes audit records $\rightarrow$ commits atomically.
- **4. Inventory Shortage Detection:** Scheduled cron or manual trigger compares current balances against minimum thresholds $\rightarrow$ Flags items where `current_quantity <= min_quantity` $\rightarrow$ Aggregates deficit metrics.
- **5. Procurement Recommendation Generation:** System evaluates stock deficits $\rightarrow$ Applies deterministic reorder formula $\rightarrow$ Gemini enriches proposal with vendor notes $\rightarrow$ Drafts procurement proposal with cost estimates.
- **6. Manager Review & Order Authorization:** Manager reviews draft proposal on dashboard $\rightarrow$ Evaluates line items and totals $\rightarrow$ Executes explicit approval or rejection with notes $\rightarrow$ Order state updates to `approved` or `rejected`.
- **7. Perishable Expiration Monitoring:** Background routine scans batch expiration dates $\rightarrow$ Filters items expiring in ≤ 7 days $\rightarrow$ Categorizes urgency $\rightarrow$ Recommends culinary specials to minimize waste.
- **8. Conversational AI Consultation:** Manager submits query via chat $\rightarrow$ Gemini agent executes registered tool calls $\rightarrow$ Service layer returns real data $\rightarrow$ Gemini formats grounded answer.
- **9. Scheduled Background Sweeps:** APScheduler fires cron job $\rightarrow$ Executes inventory health check $\rightarrow$ Evaluates shortages and near-expiry items $\rightarrow$ Records execution status to `automation_logs`.
- **10. Push Notification Alerting:** Critical stock condition detected $\rightarrow$ Formats markdown notification payload $\rightarrow$ Dispatches via Telegram Bot API asynchronously $\rightarrow$ Verifies HTTP 200 delivery.

12. API Specifications

The following catalog details the RESTful API endpoints for RestaurantOps AI. Endpoints verified in the current codebase (Stages 1–3) are designated as **Implemented**, while upcoming endpoints are marked as *Planned*.

Method	Endpoint Path	Status	Description
GET	`/api/v1/health`	**Implemented**	System health check, database ping, and environment status.
GET	`/api/v1/ingredients`	**Implemented**	List ingredients with pagination, category filter, and search.
POST	`/api/v1/ingredients`	**Implemented**	Register a new ingredient with SKU and threshold validation.
GET	`/api/v1/ingredients/{id}`	**Implemented**	Retrieve complete ingredient details by unique ID.
PUT	`/api/v1/ingredients/{id}`	**Implemented**	Update ingredient pricing, thresholds, or metadata.
DELETE	`/api/v1/ingredients/{id}`	**Implemented**	Delete ingredient (rejected if linked to recipes or sales).
GET	`/api/v1/menu-items`	**Implemented**	List all menu items with pricing and active status.
POST	`/api/v1/menu-items`	**Implemented**	Create a new restaurant menu item.
GET	`/api/v1/menu-items/{id}`	**Implemented**	Retrieve a menu item and its recipe bill-of-materials.
PUT	`/api/v1/menu-items/{id}`	**Implemented**	Update menu item details or active status flag.
POST	`/api/v1/menu-items/{id}/recipe`	**Implemented**	Define or replace recipe ingredients for a menu item.
GET	`/api/v1/menu-items/{id}/recipe`	**Implemented**	Retrieve recipe ingredient quantities for a menu item.
GET	`/api/v1/sales`	**Implemented**	Query historical sales transactions with pagination.
POST	`/api/v1/sales`	**Implemented**	Record sale and execute atomic recipe stock deductions.

GET	`/api/v1/sales/{id}`	**Implemented**	Retrieve itemized sale details and consumed recipe line items.
POST	`/api/v1/inventory/adjust`	**Implemented**	Record manual stock reconciliation, receiving, or waste.
GET	`/api/v1/inventory/transactions`	**Implemented**	Query immutable inventory movement audit log.
GET	`/api/v1/inventory/low-stock`	**Implemented**	Retrieve list of ingredients at or below minimum threshold.
GET	`/api/v1/inventory/expiring`	**Implemented**	Retrieve ingredients approaching expiration date.
GET	`/api/v1/dashboard/summary`	**Implemented**	Aggregate KPIs: stock value, low-stock count, sales totals.
POST	`/api/v1/agent/chat`	*Planned*	Conversational interface endpoint for Gemini AI assistant.
GET	`/api/v1/procurement/recommendations`	*Planned*	Fetch active AI-generated procurement proposals.
POST	`/api/v1/procurement/recommendations/{id}/approve`	*Planned*	Approve a procurement proposal (transitions to approved).
POST	`/api/v1/procurement/recommendations/{id}/reject`	*Planned*	Reject a procurement proposal with manager rationale.
GET	`/api/v1/reports/daily`	*Planned*	Generate consolidated daily operational performance report.
GET	`/api/v1/alerts`	*Planned*	Fetch active system alerts and Telegram delivery history.

13. User Interface and System Components

The frontend is engineered as a responsive Single Page Application (SPA) prioritizing clean data visualization, rapid kitchen navigation, and mobile tablet compatibility:

- **Executive KPI Dashboard:** Displays high-level operational metrics: Total Inventory Asset Value, Low Stock Count, Approaching Expirations, Today's Sales Volume, and 7-day Recharts stock consumption trends.
- **Collapsible Sidebar Navigation:** Persistent desktop and mobile sidebar providing immediate access to Pantry Inventory, Menu & Recipes, Sales Register, Procurement Workbench, AI Assistant, and System Logs.
- **Top Operational Bar:** Houses breadcrumb pathing, backend API health status pill (Green/Red), quick-action Manual Stock Adjustment launcher, and local meteorological snippet.
- **Inventory Data Grid:** Searchable, sortable, and category-filtered grid displaying current stock against minimum thresholds, unit costs, dynamic status badges (In Stock, Low Stock, Depleted), and inline editing.
- **Menu & Recipe Studio:** Interface for composing culinary items and mapping bill-of-materials ingredient weights and portion measures.
- **Sales Register Console:** Simplified POS simulation interface allowing operators to log menu dish sales and monitor automatic ingredient consumption in real time.
- **AI Assistant Drawer:** Interactive conversational terminal for natural language queries with Google Gemini, equipped with one-click quick prompt pills.
- **Procurement Review Workbench:** Review board for evaluating draft purchasing recommendations, editing proposed quantities, inspecting cost impacts, and executing manager approvals or rejections.

14. Technology Stack

Component Layer	Technology	Version	Architectural Responsibility
Frontend Framework	React	19.x	Declarative component UI architecture and virtual DOM.
Frontend Language	TypeScript	5.x	Type safety and schema contract parity with backend models.
Build Tooling	Vite	6.x	Fast ESM bundler and development proxy server.
CSS Framework	Tailwind CSS	4.x	Utility-first responsive layout and design system.

Data Visualization	Recharts	2.x	Interactive SVG consumption velocity and sales charts.
Backend Framework	FastAPI	0.115+	Asynchronous REST API controllers and dependency injection.
Language Runtime	Python	3.11+	Core server execution environment and math logic.
Data Validation	Pydantic	v2.x	Payload parsing, serialization, and input validation.
Database ORM	SQLAlchemy	2.x (Async)	Async ORM mapping, relation loading, and queries.
Database Engine	SQLite / aiosqlite	3.x	Zero-ops local relational storage with async driver.
Database Migrations	Alembic	1.14+	Declarative, version-controlled schema evolution.
AI Reasoning Engine	Google Gemini	Current	Conversational agent and structured tool calling.
AI Python SDK	google-genai	2.x	Official client library for Gemini API interactions.
Task Scheduling	APScheduler	3.x	Background cron scheduler for automated health sweeps.
Outbound Alerts	Telegram Bot API	Current	Push delivery of urgent shortage and summary messages.
Meteorological Data	Open-Meteo	Current	Contextual local temperature and precipitation feed.
Automated Testing	Pytest / Asyncio	Current	Asynchronous test suite verifying endpoints and models.

15. Minimum Viable Product (MVP) Scope

To ensure high software quality and rapid time-to-market, the MVP boundary is explicitly partitioned into included core capabilities and deliberate architectural exclusions:

Included Core Capabilities (In-Scope MVP)	Explicit Exclusions (Out-of-Scope MVP)
• Full CRUD management for pantry ingredients. • Menu item creation with multi-ingredient recipes. • Atomic sales recording with automated recipe deductions. • Negative stock prevention and validation rules. • Immutable inventory transaction audit ledger. • Deterministic low-stock and near-expiry detection. • AI-assisted procurement proposal generation. • Manager approval/rejection procurement workflow. • Conversational Gemini AI assistant with service tools. • Telegram notification alerts for critical conditions. • Contextual Open-Meteo weather integration. • Background APScheduler inventory health checks. • Executive KPI dashboard with Recharts trends.	• **No automated supplier purchasing:** System will not execute external purchase orders via vendor portals without manual approval. • **No live payment processing:** Excludes credit card gateways or physical payment hardware. • **No direct supplier EDI integration:** Electronic data interchange with wholesale distributors is reserved for future phases. • **No multi-branch enterprise support:** Scoped for single-restaurant operation; no multi-location synchronization. • **No complex ML predictive models:** Forecasting relies on deterministic run rates rather than multi-year neural nets. • **No native mobile applications:** Excludes native iOS/Android binaries; supported via responsive web.

16. Future Enhancements

Following initial deployment and operational stabilization of the MVP, the roadmap envisions several architectural expansions:

- **Predictive Machine Learning Demand Forecasting:** Incorporate neural network regression models factoring in local holidays, seasonal trends, and reservations to predict future ingredient demand weeks in advance.
- **Direct Supplier Electronic Ordering (EDI):** Connect approved purchase orders directly to major food distributors (Sysco, US Foods, Gordon Food Service) via punchout or API protocols upon manager approval.
- **Multi-Unit Enterprise Centralization:** Expand data models to support restaurant groups with shared commissary kitchens, inter-branch stock transfers, and consolidated master purchasing.
- **Camera Barcode & RFID Stock Scanning:** Enable kitchen mobile devices to scan GS1 barcodes during delivery intake, instantly reconciling physical receipts against expected purchase orders.
- **Kitchen Display System (KDS) & POS Integrations:** Direct plug-and-play webhook connectors for commercial POS platforms (Toast, Square, Clover, Aloha) to eliminate manual sales simulation entry.

- **Accounting & Invoicing Synchronization:** Automated two-way integration with general ledger accounting systems (QuickBooks Online, Xero, Sage) to synchronize actual Food Costs and invoices.

17. Acceptance Criteria

Verification of system functionality and readiness for production sign-off shall be determined against twelve concrete acceptance conditions:

Criteria ID	Verification Category	Acceptance Condition & Validation Method
AC-01	Inventory CRUD	Ingredients can be created, viewed, updated, and deleted. Submitting a duplicate SKU returns an explicit HTTP 400 Bad Request error.
AC-02	Recipe Integrity	Menu items can be mapped to multiple ingredients. Deleting an ingredient referenced by an active recipe is rejected with referential integrity failure.
AC-03	Atomic Depletion	Recording a dish sale deducts exact recipe quantities across all underlying ingredients in a single atomic transaction.
AC-04	Shortage Protection	Attempting to record a sale where any required ingredient has insufficient stock is blocked cleanly, leaving all stock levels unchanged.
AC-05	Movement Auditing	Every inventory addition, sale deduction, manual adjustment, and waste log writes an immutable record to inventory_transactions.
AC-06	Deterministic Reorders	Procurement proposals calculate suggested quantities using the exact threshold formula: $Q = \max((2 \times \text{min}) - \text{current}, \text{min})$.
AC-07	Human Approval Gate	Procurement recommendations remain in draft state until an authorized manager explicitly approves or rejects them.
AC-08	Perishable Expiry	Ingredients expiring within the configured lookahead horizon (default: 7 days) are returned accurately by the expiry monitoring endpoint.
AC-09	AI Tool Grounding	The Gemini assistant answers user questions by calling validated backend service functions; direct database querying is strictly prohibited.
AC-10	Telegram Alerts	Configured low-stock and near-expiry conditions trigger asynchronous Telegram bot notification messages to the target channel.
AC-11	Automation Auditing	APScheduler scheduled sweeps execute on schedule without user presence and record execution timestamps, status, and traces to automation_logs.
AC-12	Response Uniformity	All REST API responses conform to a unified JSON response envelope (ApiSuccess<T> or ApiError).

18. Deployment and Resource Considerations

18.1 Local Development Environment:

The project requires Python 3.11+ and Node.js 18+. Backend setup entails creating a Python virtual environment, installing dependencies from requirements.txt, and applying Alembic migrations via alembic upgrade head. Frontend setup requires installing npm packages and starting the Vite dev server with proxy routing to localhost:8000.

18.2 Server Environment Variables (backend/.env):

```
PROJECT_NAME="RestaurantOps AI"
DATABASE_URL="sqlite+aiosqlite:///./restaurantops.db"
GEMINI_API_KEY="AIzaSy..."             # Google AI Studio API Key (backend only)
TELEGRAM_BOT_TOKEN="123456:ABC..."      # Telegram BotFather Token
TELEGRAM_CHAT_ID="-100..."             # Manager Notification Channel ID
WEATHER_LATITUDE="40.7128"             # Restaurant Location Latitude (NYC)
WEATHER_LONGITUDE="-74.0060"           # Restaurant Location Longitude (NYC)
LOG_LEVEL="INFO"
```

18.3 Resource and Free-Tier Limits:

• **Google Gemini API:** Free-tier Google AI Studio API keys operate under Requests Per Minute (RPM) and Tokens Per Minute (TPM) limits. The application incorporates exponential backoff to handle transient 429 quota codes without crashing.
• **Open-Meteo API:** Operates without an API key under open-access terms for non-commercial rates. Local caching prevents redundant hourly queries.
• **Telegram Bot API:** Free of charge with a global broadcast limit of 30 messages/second, well within single-restaurant needs.

19. Conclusion

RestaurantOps AI bridges a critical operational gap in modern hospitality management by fusing deterministic, recipe-driven inventory math with advanced generative reasoning powered by Google's Gemini API. By automating routine consumption deductions, detecting low-stock emergencies prior to peak service, and actively monitoring perishable expiration horizons, the system directly tackles the core drivers of food cost inflation and kitchen chaos.

With a steadfast architectural commitment to a unified service layer, complete transaction auditability, strict secret isolation, and human-in-the-loop purchasing governance, RestaurantOps AI establishes a production-grade software foundation designed to optimize restaurant profitability today while paving the way for enterprise hospitality expansion.